QNTSim: Quantum Network Simulator

A powerful conceptual and teaching tool for the next generation of quantum engineers and physicists.

QNTSim is the first quantum network simulation platform in India built for academia and research training. It helps students, instructors and laboratories visualise quantum phenomena such as entanglement and quantum communication through an intuitive, user-guided learning environment, with customisable network setups to explore, test and understand quantum communication protocols.

Every student builds a quantum network

No optical bench, no photon source, no queue for laboratory time.

Architecturally, QNTSim layers Access & Control (GUI and API) over a discrete-event engine handling time-ordered event scheduling and resource management, which drives an application and protocol logic layer — high-level protocols, services and logic such as QKD networks and teleportation — operating over quantum system models covering abstracted devices, channels and noise. Network elements modelled include repeater nodes with swapping and purification, BSM nodes, classical and quantum channels, and quantum storage.

Key Features

- **Intuitive GUI:** Drag-and-drop network design with configurable control panels for parameter tuning and reruns; simulate quantum communication workflows and QKD protocols with adjustable network parameters.
- **Entanglement Engine:** Simulate entanglement generation, purification and swapping with meet-in-the-middle BSM nodes.
- **Physics-Grounded Outputs:** Observe the impact of coherence time, detector efficiency and fidelity degradation due to noise on entanglement distribution and protocol-level performance.
- **Configurable Learning Tool:** Protocol-specific simulation with overview, statistics, inferences and sweep mode.
- **User-Adjustable Parameters:** Realistic quantum network behaviour through tunable attenuation, detector efficiency and memory fidelity.

Technical Specifications

Configurable Learning Tool	Protocol-specific: overview, statistics, inferences and sweep mode
Supported Protocols	E2E, CHSH test, E91, BBM92
Entanglement Support	Entanglement generation (Barrett-Kok / single heralded), entanglement purification (BBPSSW), entanglement swapping
Network Topologies	Customisable, up to 20 nodes
Physical Layer Models	Quantum and classical channels, detectors, sources, memories
Tunable Parameters	Channel length and attenuation, detector efficiency, memory fidelity, protocol choice
Simulation Type	Single run simulation, parameter sweep simulation, batch simulation
Architecture	Modular, controller-driven simulation engine
Output	Protocol outputs (entangled pairs, S value, key metrics), execution and simulation time, and fidelity tracking
Operating Systems	All major operating systems like macOS, Windows, Linux and Arch-based
Deployment	Cloud, on-premises or application-based

Learning Outcomes

Students will learn:

- BB84, E91, BBM92 and B92 protocol simulation, spanning non-orthogonal-state and entanglement-based QKD approaches
- Bell state generation and measurement, observing quantum correlations and CHSH-inequality violations
- Realistic entanglement generation via the Barrett-Kok and single heralded protocols, followed by entanglement purification (fidelity gain over repeated rounds) and entanglement swapping across distant nodes, the core principle behind quantum repeaters
- Quantum teleportation and multi-node GHZ-state distribution, analysing fidelity and multi-partite entanglement properties
- Basic BB84, E91 and BBM92 protocol simulation is accessible at undergraduate level. Entanglement generation (Barrett-Kok and single heralded), purification (BBPSSW) and swapping are postgraduate and research-level network-security topics.

Experiments Included

Nine experiments, from undergraduate protocol simulation through to postgraduate and research-level entanglement engineering.

Exp. no	Experiment Name	Description
1	**BB84 Protocol**	Simulate the first QKD protocol using non-orthogonal quantum states and observe secure key generation and error detection under noise or eavesdropping
2	**E91 Protocol**	Explore entanglement-based QKD and verify quantum correlations through Bell test violations while generating shared keys
3	**BBM92 Protocol**	Analyse this entanglement-based variation of BB84 using fewer quantum bases, focusing on security and key extraction efficiency
4	**B92 Protocol**	Study QKD with only two non-orthogonal states and observe the trade-off between security and error sensitivity
5	**Bell State Generation & Bell Measurements**	Create and measure entangled Bell states; observe quantum correlations, measurement collapse and CHSH inequality violations
6	**Entanglement Generation (Barrett-Kok Protocol)**	Simulate realistic entanglement creation using matter qubits and optical interference; study fidelity loss and success conditions
7	**Entanglement Purification**	Use BBPSSW to improve the quality of noisy entangled states by simulating purification steps; observe fidelity gain over repeated rounds
8	**Entanglement Swapping**	Demonstrate how entanglement can be extended over distant nodes without direct interaction, a key principle in quantum repeaters
9	**Quantum Teleportation**	Transfer an unknown quantum state using entanglement and classical communication; observe fidelity and qubit reconstruction

Skills Gained

Students will gain:

- Network protocol modelling for quantum key distribution
- Entanglement engineering; purification, swapping and teleportation
- Quantum-secure communication protocol design

Industry Mapping

- Quantum-repeater R&D and national quantum network programmes
- Academia and research like metropolitan quantum-internet testbeds
- Cybersecurity consulting and defence secure-communications planning
- Network-simulation software industry and R&D tooling
- Telecom quantum-network infrastructure and quantum-internet planning

Quantum networking is described in current industry analysis as moving from laboratory demonstration toward early infrastructure. National security agencies are typically the earliest adopters, major telecom and utility players are investing in network infrastructure, and metropolitan quantum-internet testbeds connecting a handful of nodes are anticipated within the next decade, built on precisely the protocol stack (BB84, E91, BBM92, repeater swapping and purification) that QNTSim simulates.

Curriculum Mapping

How QNTSim Enables the Key Learning Outcomes	Course Code	AICTE Course Title
Overview of the field of quantum technologies, familiarising students with the state of the art across all four verticals	**QT01**	Survey of Quantum Technologies and Applications
Polarisation optics; linear and square-law detectors; information theory; basics of quantum information; components of quantum networks; teleportation, dense coding and QKD protocols; survey of hardware implementations	**QTM 11**	Introduction to Quantum Communication
Topics in electrical, electronics and communication engineering, and in computer science, relevant to quantum computation, communication and sensing	**QT09**	Engineering Foundations of Quantum Technologies
Polarisation optics; linear and square-law detectors; information theory; basics of quantum information; components of quantum networks; teleportation, superdense coding and QKD protocols; survey of hardware implementations	**QT06**	Introduction to Quantum Communication

QTXX refers to the AICTE Minor Degree for UG programmes in Quantum Technologies; QTMXX refers to the AICTE M.Tech Programme in Quantum Technologies.

Pre-requisites

- Basics of quantum information
- Linear algebra
- Polarisation optics — optional; covered in the QNTSim learning module

Licensing Model

Software licence, applicable when the software is deployed on a dedicated in-house server for use by multiple users or systems.

Requirements

QNTSim Server for up to 30 concurrent users

Specification	Minimum Requirements	Recommended
CPU	16-core modern processor	24–32 high-performance cores
RAM	32 GB	64 GB
Storage	500 GB NVMe SSD	1 TB NVMe
Network	1 Gbps	1–10 Gbps
Operating System	Ubuntu Server 24.04 LTS 64-bit	Ubuntu Server 24.04 LTS 64-bit
Topology	15 nodes	25 nodes
Users	Up to 30 concurrent users	50 concurrent users

No optical table, cleanroom or laboratory infrastructure is required for this platform.

What's Included

Learning Module	Simulation Playground	Learning Management System
Quantum Communication Fundamentals	Single Simulation	Batch Creation
Quantum Hardware Concepts	Simulation Comparison	Professor Management
Experimental Setup Learning	Sweep Simulation	Student Management
Protocol Theory	Timeline Visualisation	Learning Analytics
Interactive Visualisations	Event Visualisation	Simulation Analytics
Assessments & Quizzes	Simulation Export	Student Progress Reports
Learning Progress Tracking	Result Sharing	Batch Comparison
Certification	Saved Simulation Templates	Department Analytics
		Institution Dashboard

Frequently Asked Questions

What computer, operating system or software is needed to operate it?

QNTSim runs on all major operating systems like macOS, Windows, Linux and Arch-based systems.

What is the licensing model: single-seat or lab-wide?

Both. The licence model follows the deployment mode: single-seat application-based licences, and multi-seat cloud deployment where user numbers are set by plan.

Is this cloud-hosted or on-premises?

Either. Deployment can be on cloud, on premises, or application-based.

What is the warranty period, and is an AMC or support plan available?

Updates are provided over the period of the engagement. Where a bug or issue is found, maintenance updates are issued as agreed.

Is on-site installation and faculty or student training provided?

Yes.

Is the software suitable for UG, PG or PhD-level work?

Yes. Basic BB84, E91 and BBM92 protocol simulation is accessible at undergraduate level, while entanglement generation (Barrett-Kok and single heralded), purification (BBPSSW) and swapping are postgraduate and research-level network-security topics.

Can experiments or protocols be customised or extended by faculty?

Yes. QNTSim is explicitly designed to let users design and build novel quantum network protocols, not only run the built-in catalogue — with customisable network topologies of up to 15 nodes and tunable physical-layer parameters including channel attenuation, detector efficiency and memory fidelity.